

2

Infinite Series

Motivated by (1.15) at the end of the previous chapter, we wish to define an infinite sum of power functions; however, before we can do so, we need to define an infinite sum of real (or complex) numbers. That is, our first task will be to consider an infinite sequence of numbers $a_1, a_2, \dots, a_n, \dots$, and examine when and how we can make sense of the sum

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots .$$

The formal definition of adding infinitely many numbers will be given in Section 2.1. Compared with addition of finitely many numbers, it turns out that many complications appear. For example, the sum might depend on the way the numbers are ordered; and in many cases, only approximative values for the sum can be found. Section 2.1 also introduces some of the criteria which are used to decide whether a given infinite summation makes sense. Section 2.2 describes some methods for estimating infinite sums. After these preliminary steps, we turn to a subject which will follow us during the entire book, namely, infinite sums with terms depending on a variable. We analyze such cases with increasing complexity: in Section 2.3 we consider geometric series, i.e., sums of the type (x denotes the variable)

$$1 + x + x^2 + \cdots + x^n + \cdots . \tag{2.1}$$

In Section 2.4 we consider “similar” series, where we allow the terms in (2.1) to be multiplied by some constants, i.e., the series are of the form

$$a_0 + a_1x + a_2x^2 + \cdots + a_nx^n + \cdots$$

for certain constants a_n ; series of this type are called *power series*. Section 2.5 goes one important step further: we consider infinite sums of general functions, defined on some interval. An understanding of this subject is instrumental for the analysis of Fourier series and wavelets. Section 2.6 introduces uniform convergence, and finally, Section 2.7 gives an idea about how infinite series can be useful for signal transmission.

In the entire chapter we focus on the approximation properties related to infinite series. That is, given an infinite series of numbers, how can we estimate how many terms we need to include in order to obtain a realistic approximation of the sum? And given an infinite series representing a function, how can we guarantee that the finite partial sums share the properties of the function?

2.1 Infinite series of numbers

We begin by defining an infinite sum of real (or complex) numbers $a_1, a_2, a_3, \dots, a_n, \dots$.

Formally, an *infinite series* is an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots = \sum_{n=1}^{\infty} a_n.$$

The numbers a_n are called the *terms* in the infinite series. In order to explain how an infinite series should be interpreted, we first define the N th *partial sum* associated to the infinite series as

$$S_N := a_1 + a_2 + a_3 + \cdots + a_N = \sum_{n=1}^N a_n. \quad (2.2)$$

Definition 2.1.1 Let $\sum_{n=1}^{\infty} a_n$ be an infinite series with partial sums S_N . If S_N is convergent, i.e., if there exists a number S such that

$$S_N \rightarrow S \text{ as } N \rightarrow \infty, \quad (2.3)$$

we say that the infinite series $\sum_{n=1}^{\infty} a_n$ is convergent with sum S . We write this as

$$\sum_{n=1}^{\infty} a_n = S.$$

In case S_N is divergent, $\sum_{n=1}^{\infty} a_n$ is said to be divergent. No sum is attributed to a divergent series.

Note that if we express the definition directly via the terms in the infinite series, to say that $\sum_{n=1}^{\infty} a_n$ is convergent with sum S means exactly that

$$\sum_{n=1}^N a_n \rightarrow S \text{ as } N \rightarrow \infty;$$

thus, to consider an infinite sum really corresponds to the intuitive idea of just “adding more and more terms”.

To avoid confusion, we mention that sometimes we will consider infinite series starting with a_0 instead of a_1 . The difference between $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=0}^{\infty} a_n$ is that in the second case we have added the number a_0 ; so for the question whether the series is convergent or not, it does not matter whether we start with $n = 0$ or $n = 1$. In case of convergence,

$$\sum_{n=0}^{\infty} a_n = a_0 + \sum_{n=1}^{\infty} a_n.$$

To determine whether a series is convergent or not, one usually applies the so-called convergence criteria, to which we return later in this section. In some cases we can apply the definition directly:

Example 2.1.2 In this example we consider the infinite series

- (i) $\sum_{n=1}^{\infty} n = 1 + 2 + \cdots + n + \cdots$;
- (ii) $\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^n} + \cdots$;
- (iii) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} + \cdots$.

For the infinite series in (i), the N th partial sum is

$$S_N = \sum_{n=1}^N n = 1 + 2 + \cdots + N.$$

It is clear that

$$S_N \rightarrow \infty \text{ for } N \rightarrow \infty$$

(in fact, $S_N = \frac{N(N+1)}{2}$ for all $N \in \mathbb{N}$). Thus

$$\sum_{n=1}^{\infty} n \text{ is divergent.}$$

Now we consider the infinite series in (ii). The N th partial sum is

$$S_N = \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^N} = \sum_{n=1}^N \frac{1}{2^n}.$$

We now give a geometric argument showing that

$$\sum_{n=1}^N \frac{1}{2^n} = 1 - \frac{1}{2^N}. \quad (2.4)$$

In fact, $S_1 = \frac{1}{2}$, which is exactly half of the distance from 0 to 1. That is,

$$S_1 = 1 - \frac{1}{2}.$$

Further, $S_2 = \frac{1}{2} + \frac{1}{2^2}$; thus, the extra contribution compared to S_1 is $\frac{1}{2^2}$, which is half the distance from S_1 to 1. Going to the next step, i.e., looking at S_3 , again cuts the distance to 1 by a factor of 2:

$$S_3 = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} = 1 - \frac{1}{2^3}.$$

The same happens in all the following steps: when looking at S_N , i.e., after N steps, the distance to 1 will be exactly $1/2^N$, which shows (2.4). It follows that

$$\sum_{n=1}^N \frac{1}{2^n} \rightarrow 1 \text{ for } N \rightarrow \infty;$$

we conclude that

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \text{ is convergent with sum } 1.$$

Concerning the series in (iii), we first observe that for any $n \in \mathbb{N}$,

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1};$$

thus, the N th partial sum is

$$\begin{aligned} S_N &= \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{N(N+1)} \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{N} - \frac{1}{N+1}\right) \\ &= 1 - \frac{1}{N+1}. \end{aligned}$$

Since $S_N \rightarrow 1$ as $N \rightarrow \infty$, we conclude that the series is convergent with sum 1. $\square$

Intuitively, the “problem” with the series in Example 2.1.2 (i) is that the terms $a_n = n$ are too large: considering partial sums S_N for larger and larger values of N means that we continue to add numbers to S_N , and when they grow as in this example, it prevents S_N from having a finite limit as $N \rightarrow \infty$. A refinement of this intuitive statement leads to a useful criterion,

which states that an infinite series only has a chance to be convergent if $a_n \rightarrow 0$ as $n \rightarrow \infty$; this result is called the *n th term test*.

Theorem 2.1.3 *If $a_n \not\rightarrow 0$ as $n \rightarrow \infty$, then $\sum_{n=1}^{\infty} a_n$ is divergent.*

Proof: Assume that $\sum_{n=1}^{\infty} a_n$ is convergent with sum S . By definition of the partial sums, this means that

$$S_N = a_0 + a_1 + \cdots + a_N \rightarrow S \text{ as } N \rightarrow \infty;$$

this implies that

$$a_N = S_N - S_{N-1} \rightarrow S - S = 0 \text{ as } N \rightarrow \infty. \quad \square$$

An alternative (but equivalent) formulation of Theorem 2.1.3 says that

if $\sum_{n=1}^{\infty} a_n$ is convergent, then $a_n \rightarrow 0$ for $n \rightarrow \infty$.

It is important to notice that Theorem 2.1.3 (at most) can give us a negative conclusion: we can *never* use the n th term test to conclude that a series is convergent. In fact, even if $a_n \rightarrow 0$ as $n \rightarrow \infty$, it can happen that $\sum_{n=1}^{\infty} a_n$ is divergent; see Example 2.2.3.

Intuitively, it is very reasonable that if a certain infinite series with positive terms is convergent, then any series with smaller (positive) terms is also convergent; in fact, the partial sums of the series with the smaller terms are forced to stay bounded. The formal statement of this result is called the *comparison test*:

Theorem 2.1.4 *Assume that $0 \leq a_n \leq b_n$ for all $n \in \mathbb{N}$. Then the following holds:*

- (i) *If $\sum_{n=1}^{\infty} b_n$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent as well.*
- (ii) *If $\sum_{n=1}^{\infty} a_n$ is divergent, then $\sum_{n=1}^{\infty} b_n$ is divergent as well.*

Example 2.1.5 Consider the infinite series

$$\sum_{n=1}^{\infty} \frac{1}{(n+1)^2} = \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{(n+1)^2} + \cdots$$

First, observe that for all $n \in \mathbb{N}$,

$$\frac{1}{(n+1)^2} \leq \frac{1}{(n+1)n}.$$

By Example 2.1.2(iii) we know that $\sum_{n=1}^{\infty} \frac{1}{(n+1)n}$ is convergent, so the comparison criterion shows that $\sum_{n=1}^{\infty} \frac{1}{(n+1)^2}$ is convergent. This coincides with a result we will obtain much later in Example 3.6.2, with a very different argument which even allows us to find the sum. $\square$

For the question whether a given series $\sum_{n=1}^{\infty} a_n$ is convergent or not, the first terms in the series do not matter: the question is whether a_n converges sufficiently fast to zero as $n \rightarrow \infty$. It turns out that if two series “behave similarly” for large values of n , they are either both convergent, or both divergent. In order to state this result formally, we need a definition: we say that two series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ with positive terms are *equivalent* if there exists a constant $C > 0$ such that

$$\frac{a_n}{b_n} \rightarrow C \text{ for } n \rightarrow \infty.$$

Proposition 2.1.6 *Assume that the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ have positive terms and are equivalent. Then $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are either both convergent, or both divergent.*

Proposition 2.1.6 gives an alternative argument for the result in Example 2.1.5: the reader can check that the series $\sum_{n=1}^{\infty} \frac{1}{(n+1)n}$ and $\sum_{n=1}^{\infty} \frac{1}{(n+1)^2}$ are equivalent.

Given a sequence of numbers, we note that our definition of the associated infinite series depends on how the terms are ordered: different orderings will lead to different partial sums, and (potentially) different answers to the question whether the series is convergent or not. This is not merely a theoretical concern, but a problem that appears in practice: in fact, there exist sequences where certain orderings lead to convergent series, and other lead to divergent series (see below). The following definition introduces a condition on the terms in an infinite series, which implies that the series is convergent with respect to any ordering.

Definition 2.1.7 *An infinite series $\sum_{n=1}^{\infty} a_n$ is said to be absolutely convergent if $\sum_{n=1}^{\infty} |a_n|$ is convergent.*

Proposition 2.1.8 *If $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent; furthermore, any reordering of the terms in the series will lead to a convergent series having the same sum.*

A series which is convergent no matter how the terms are reordered, is said to be *unconditionally convergent*. Using this terminology, Proposition 2.1.8 says that an absolutely convergent series is unconditionally convergent.

There exist series which are convergent, but not absolutely convergent; they are said to be *conditionally convergent*. For conditionally convergent series with real terms, something very strange happens. In fact, the terms can be reordered such that we obtain a divergent series; and for *any* given real number $\tilde{S}$ we can find an ordering so that the corresponding infinite series is convergent with sum equal to $\tilde{S}$. Example 2.2.5 exhibits an infinite series which is convergent, but not absolutely convergent.

A sufficient condition for (absolute) convergence of an infinite series is given by the *quotient test*:

Theorem 2.1.9 Assume that $a_n \neq 0$ for all $n \in \mathbb{N}$ and that

$$\left| \frac{a_{n+1}}{a_n} \right| \rightarrow C \in [0, \infty] \text{ for } n \rightarrow \infty. \quad (2.5)$$

Then the following hold:

- (i) If $C < 1$, then $\sum a_n$ is absolutely convergent.
- (ii) If $C > 1$, then $\sum a_n$ is divergent.

We prove Theorem 2.1.9 in Section A.4. Note that the case where (2.5) is satisfied with $C = 1$ has a very special status: Theorem 2.1.9 does not give any conclusion in that case. Example 2.2.3 will show us that there exist convergent as well as divergent series which lead to $C = 1$; so if we end up with this particular value, we have to use other methods to find out whether the given series is convergent or not.

Example 2.1.10 Consider the infinite series

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} = \frac{1}{2} + 1 + \frac{9}{8} + \cdots + \frac{n^2}{2^n} + \cdots.$$

Letting $a_n = \frac{n^2}{2^n}$, we see that

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2} = \left(1 + \frac{1}{n}\right)^2 \frac{1}{2} \rightarrow \frac{1}{2} \text{ for } n \rightarrow \infty.$$

Thus the quotient test shows that the series is convergent. $\square$

2.2 Estimating the sum of an infinite series

The definition of an infinite series involves the sum of the series. There exist several ways to check whether a series is convergent or not, but only in relatively rare cases is it possible to find the exact sum of an infinite series. This raises the question how one can find approximative values for the infinite sum. Using computers, we are able to calculate the partial sums S_N for large values of $N \in \mathbb{N}$; but in order for this information to be useful, we need to be able to estimate the difference

$$\begin{aligned} \left| \sum_{n=1}^{\infty} a_n - S_N \right| &= \left| \sum_{n=1}^{\infty} a_n - \sum_{n=1}^N a_n \right| \\ &= \left| \sum_{n=N+1}^{\infty} a_n \right|. \end{aligned}$$

One important result which can frequently be used in order to do so can be derived from the following result, called the *integral test*:

Theorem 2.2.1 *Assume that $f : [1, \infty[\rightarrow [0, \infty[$ is continuous and decreasing. Then the following holds:*

(i) *If $\int_1^t f(x)dx$ has a finite limit as $t \rightarrow \infty$, then $\sum_{n=1}^{\infty} f(n)$ is convergent, and*

$$\int_1^{\infty} f(x)dx < \sum_{n=1}^{\infty} f(n) < \int_1^{\infty} f(x)dx + f(1).$$

(ii) *If $\int_1^t f(x)dx \rightarrow \infty$ as $t \rightarrow \infty$, then $\sum_{n=1}^{\infty} f(n)$ is divergent.*

The theorem is proved in Section A.4. A reader who checks the proof will observe that it does not matter that the sum appearing in the integral test starts with $n = 1$; the same argument works starting with any positive integer-value of n . One way of formulating this observation is that if the conditions in Theorem 2.2.1 are satisfied, then for any $p \in \mathbb{N}$,

$$\int_p^{\infty} f(x)dx < \sum_{n=p}^{\infty} f(n) < \int_p^{\infty} f(x)dx + f(p). \quad (2.6)$$

We also notice that this result is obtained, even if f only satisfies the conditions in Theorem 2.2.1 starting with $x = p$. This leads to the following way of estimating the infinite sum $\sum_{n=1}^{\infty} f(n)$:

Corollary 2.2.2 *Assume that the function $f : [1, \infty[\rightarrow [0, \infty[$ is continuous and decreasing, at least from $x = p$ for some $p \in \mathbb{N}$. Then, assuming that $\int_p^t f(x)dx$ has a finite limit as $t \rightarrow \infty$,*

$$\left| \sum_{n=1}^{\infty} f(n) - \sum_{n=1}^N f(n) \right| \leq \int_{N+1}^{\infty} f(x)dx + f(N+1)$$

for all integers $N \geq p$.

Corollary 2.2.2 is proved using that

$$\sum_{n=1}^{\infty} f(n) - \sum_{n=1}^N f(n) = \sum_{n=N+1}^{\infty} f(n),$$

followed by an application of (2.6) with $p = N+1$. In the following example, this result is used to estimate an infinite sum:

Example 2.2.3 Let $\alpha \geq 1$, and consider the function

$$f(x) = \frac{1}{x^\alpha}, \quad x \in [1, \infty[.$$

This function is decreasing and continuous. If $\alpha > 1$, then

$$\int_1^t \frac{1}{x^\alpha} dx = \frac{1}{-\alpha+1} [x^{-\alpha+1}]_1^t = \frac{1}{-\alpha+1} (t^{-\alpha+1} - 1).$$

It follows that

$$\int_1^t \frac{1}{x^\alpha} dx \rightarrow \frac{1}{\alpha-1} \text{ for } t \rightarrow \infty.$$

Via Theorem 2.2.1 we conclude that

$$\sum_{n=1}^{\infty} \frac{1}{n^\alpha} \text{ is convergent if } \alpha > 1.$$

For $\alpha = 1$,

$$\int_1^t \frac{1}{x} dx = [\ln x]_1^t = \ln t \rightarrow \infty \text{ for } t \rightarrow \infty.$$

Again via Theorem 2.2.1 we conclude that

$$\sum_{n=1}^{\infty} \frac{1}{n} \text{ is divergent.}$$

Let us now find an estimate for the infinite sum $\sum_{n=1}^{\infty} \frac{1}{n^2}$, with an error of at most $\epsilon = 0.1$. To do so, we use Corollary 2.2.2, which shows that for any $N \in \mathbb{N}$,

$$\begin{aligned} \left| \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=1}^N \frac{1}{n^2} \right| &\leq \int_{N+1}^{\infty} \frac{1}{x^2} dx + \frac{1}{(N+1)^2} \\ &= \frac{1}{N+1} + \frac{1}{(N+1)^2}. \end{aligned} \quad (2.7)$$

The term in (2.7) is smaller than 0.1 already for $N = 10$; this shows that with an error of maximally 0.1,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \approx \sum_{n=1}^{10} \frac{1}{n^2} = 1.55.$$

We shall in Example 3.6.2 prove that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \approx 1.645. \quad \square$$

Corollary 2.2.2 only applies to series with positive terms. Another case, where an estimate for how well the finite partial sums approximate an infinite sum can be given, is for *alternating series*; by this, we mean a

series which can be written in the form

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - \cdots + (-1)^{n-1} b_n + \cdots, \quad (2.8)$$

where

$$b_n \geq 0 \text{ for all } n \in \mathbb{N}, \text{ or } b_n \leq 0 \text{ for all } n \in \mathbb{N}.$$

The following criterion is useful in order to check convergence for an alternating series. It can also be used to find an estimate for how many terms we need to include in the partial sum in order to reach a certain approximation of the sum:

Proposition 2.2.4 *An alternating series for which the sequence $\{b_n\}_{n=1}^{\infty}$ decreases monotonically to zero, is convergent, and for all $N \in \mathbb{N}$,*

$$\left| \sum_{n=1}^{\infty} (-1)^{n-1} b_n - \sum_{n=1}^N (-1)^{n-1} b_n \right| \leq b_{N+1}.$$

Example 2.2.5 By Proposition 2.2.4, the series

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n}$$

is convergent. According to Example 2.2.3, the series is not absolutely convergent. As already mentioned, this implies that for any given number $\tilde{S}$ we can reorder the infinite series so that we obtain a convergent series with sum equal to $\tilde{S}$. Furthermore, it can be reordered so we obtain a divergent series. $\square$

Corollary 2.2.2 and Proposition 2.2.4 are efficient tools to approximate infinite sums, whenever they apply. For general series with real terms it could be tempting to try to reorder the series in such a way that these results could be applied; however, as we have seen, this is only allowed for unconditionally convergent series.

2.3 Geometric series

The next step is to consider series where the terms depend on a variable. One of the simplest examples is a *geometric series*; by this we mean a series of the form

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots + x^n + \cdots, \quad (2.9)$$

where $x \in \mathbb{R}$ (or more generally, $x \in \mathbb{C}$). When dealing with a specific value

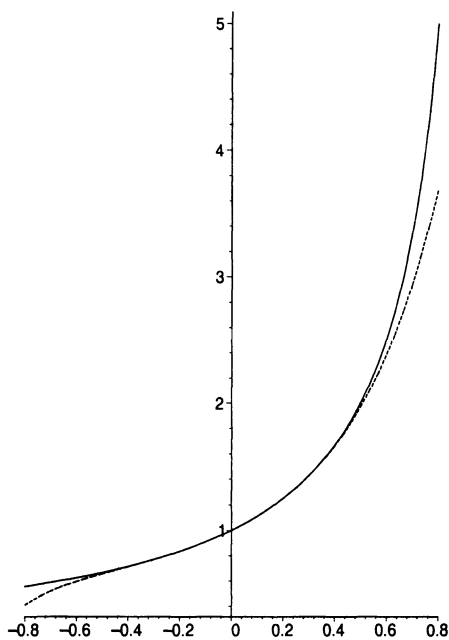


Figure 2.3.1 The function $f(x) = \frac{1}{1-x}$ and the partial sum $S_5(x) = \sum_{n=0}^5 x^n$ of the series representation in (2.10) (dotted).

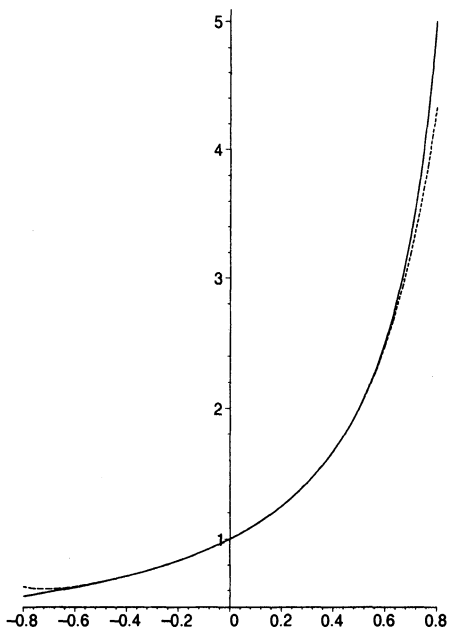


Figure 2.3.2 The function $f(x) = \frac{1}{1-x}$ and the partial sum $S_8(x) = \sum_{n=0}^8 x^n$ (dotted).

of x , this number is called the *quotient*. We now show that a geometric series is convergent if and only if $|x| < 1$:

Theorem 2.3.3 *A geometric series with quotient x is convergent if and only if $|x| < 1$; and for $|x| < 1$ the sum is*

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}. \quad (2.10)$$

Proof: First we find the values of x for which the series is convergent. For this purpose we notice that for any $N \in \mathbb{N}$,

$$\begin{aligned} (1-x)(1+x+x^2+\cdots+x^N) &= 1-x+x-x^2+\cdots+x^N-x^{N+1} \\ &= 1-x^{N+1}; \end{aligned}$$

thus, for $x \neq 1$, the N th partial sum of (2.9) is given by

$$S_N = 1+x+\cdots+x^N = \frac{1-x^{N+1}}{1-x}.$$

When $|x| < 1$, we see that $x^{N+1} \rightarrow 0$ for $N \rightarrow \infty$; therefore S_N converges for $N \rightarrow \infty$ and the limit is $\frac{1}{1-x}$. When $|x| \geq 1$, S_N does not have a limit for $N \rightarrow \infty$ (nor in the case $x = 1$ which was excluded in the above discussion). $\square$

Example 2.3.4 We consider Example 2.1.2(ii) again. The series

$$\sum_{n=1}^{\infty} \frac{1}{2^n}$$

is a geometric series with $x = \frac{1}{2}$; thus, according to Theorem 2.3.3, the series is convergent with sum $\frac{1}{1-\frac{1}{2}} - 1 = 1$. $\square$

Note that we in Example 2.3.4 found the sum of a geometric series starting by $n = 1$ by subtraction of the first term in (2.10). Alternatively, we could have used that for all $N \in \mathbb{N}$ and all $x \in]-1, 1[$,

$$\sum_{n=N}^{\infty} x^n = \sum_{n=0}^{\infty} x^{n+N} = \frac{x^N}{1-x}. \quad (2.11)$$

For reasons that will be clear in Section 2.6, we emphasize that (2.10) is a *pointwise* identity: it states, that when we *fix* a certain $x \in]-1, 1[$, then the series in (2.10) converges to the stated limit. This *does not* imply that we can find a partial sum S_N of (2.9) which is close to $\frac{1}{1-x}$ for *all* $x \in]-1, 1[$; that is, the partial sums do not approximate $\frac{1}{1-x}$ in the uniform sense of (1.2). See Figures 2.3.1–2.3.2; we return to the question of uniform approximation in Section 2.6.

2.4 Power series

We are now ready to connect the infinite series with the approximation theory discussed in Chapter 1. We begin with a reformulation of Theorem 1.3.6 using infinite series:

Theorem 2.4.1 *Let I be an interval and f an arbitrarily often differentiable function defined on I . Let $x_0 \in I$, and assume that there exists a constant $C > 0$ such that*

$$|f^{(n)}(x)| \leq C \quad \text{for all } n \in \mathbb{N} \quad \text{and all } x \in I.$$

Then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \quad \text{for all } x \in I. \quad (2.12)$$

Proof: The statement in Theorem 1.3.6 shows that the N th partial sum of the infinite series in (2.12) converges to $f(x)$ when $N \rightarrow \infty$; thus, the result follows by the definition of an infinite sum. $\square$

In the remainder of this chapter we will concentrate on the case where $0 \in I$ and we can choose $x_0 = 0$.

Example 2.4.2 Let us go back to Example 1.3.3, where we considered the function

$$f(x) = e^x, \quad x \in]-1, 2[.$$

Since

$$f'(x) = f''(x) = \cdots = e^x,$$

we see that

$$|f^{(n)}(x)| \leq e^2 \quad \text{for all } n \in \mathbb{N}, \quad x \in]-1, 2[. \quad (2.13)$$

Thus, according to Theorem 2.4.1,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n, \quad x \in]-1, 2[. \quad (2.14)$$

The particular interval $] -1, 2[$ does not play any role in (2.14): the arguments leading to this identity can be performed for any bounded interval containing 0. This implies that the identity holds pointwise for *any* $x \in \mathbb{R}$. Using that $f^{(n)}(0) = 1$ for all $n \in \mathbb{N}$, we arrive at

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad x \in \mathbb{R}. \quad (2.15)$$

Formula (2.15) shows that calculation of e^x can be reduced to repeated use of the operations addition, multiplication, and division. All calculators (and mathematical calculating programs like Maple, Mathematica or Matlab) are based only on these operations (together with subtraction) and all other computations are done by reduction to these. In fact, calculators evaluate expressions involving e^x using series expansions like the one in (2.15)! However, a mathematical program can not add infinitely many terms in finite time, so in practice it calculates

$$\sum_{n=0}^N \frac{x^n}{n!} \approx e^x$$

for a large value of N ; see Example 1.3.7, where we saw that $N = 8$ was sufficient to approximate e^x with an error of at most 0.015 on the interval $[-1, 2]$. The commercial mathematical programs use way more terms, and the magnitude of the round-off error is considerably smaller; under “normal” conditions one will not notice the round-off error, but sometimes with long calculations the errors add up and become noticeable. $\square$

Via the result in (2.15) we can now find an exact value for the integral considered in Example 1.3.8, in terms of an infinite series:

Example 2.4.3 In Example 1.3.8, we gave an estimate for the integral

$$\int_0^1 e^{-x^2} dx.$$

Now, by (2.15),

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!}, \quad x \in \mathbb{R}.$$

One can prove that it is allowed to integrate a power series over a bounded interval term-wise, see Corollary 2.6.8 ; in the present case, this shows that

$$\begin{aligned} \int_0^1 e^{-x^2} dx &= \int_0^1 \left(\sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} \right) dx \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^1 x^{2n} dx \\ &= \left[\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)n!} \right]_{x=0}^1 \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)n!}. \end{aligned} \tag{2.16}$$

Thus, we have now obtained an exact expression for the integral in terms of an infinite series. In contrast, Example 1.3.8 gave an approximative value. We can obtain an arbitrarily precise value for the integral, simply by including sufficiently many terms in the partial sum of (2.16). We can even estimate how many terms we need to include in order to obtain a certain precision: according to Proposition 2.2.4,

$$\left| \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)n!} - \sum_{n=0}^N (-1)^n \frac{1}{(2n+1)n!} \right| \leq \frac{1}{(2N+3)(N+1)!}.$$

This result shows that $N = 8$ leads to a better approximation of the integral than suggested by Exercise 1.3; see Exercise 2.6. $\square$

Example 2.4.4 Consider the function $f(x) = \sin x$, $x \in \mathbb{R}$. Then

$$\begin{aligned} f'(x) &= \cos x = f^{(1+4n)}(x), \text{ and } f^{(1+4n)}(0) = 1, \quad n \in \mathbb{N}, \\ f''(x) &= -\sin x = f^{(2+4n)}(x), \text{ and } f^{(2+4n)}(0) = 0, \quad n \in \mathbb{N}, \\ f^{(3)}(x) &= -\cos x = f^{(3+4n)}(x), \text{ and } f^{(3+4n)}(0) = -1, \quad n \in \mathbb{N}, \\ f^{(4)}(x) &= \sin x = f^{(4n)}(x), \text{ and } f^{(4n)}(0) = 0, \quad n \in \mathbb{N}. \end{aligned}$$

We see that $|f^{(n)}(x)| \leq 1$ for all $n \in \mathbb{N}$ and all $x \in \mathbb{R}$. By Theorem 2.4.1,

$$\begin{aligned} \sin x &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \\ &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}. \end{aligned} \tag{2.17}$$

Our calculators find $\sin x$ using this expression. The partial sums of (2.17) converge very fast to the sine function, mainly because of the denominator $(2n+1)!$ in (2.17), which makes the coefficient in the power series decay quickly. Another reason for the fast convergence is the oscillatory behavior of the terms in the partial sums: the series (2.17) is an alternating series, and according to Proposition 2.2.4,

$$\left| \sin x - \sum_{n=0}^N \frac{(-1)^n}{(2n+1)!} x^{2n+1} \right| \leq \frac{|x|^{2N+3}}{(2N+3)!}. \tag{2.18}$$

If we want the difference in (2.18) to be smaller than 0.01 for all $x \in [-5, 5]$, it is enough to take $N \geq 7$; for $N = 7$, the corresponding partial sum is a polynomial of degree 15.

It is actually surprising that the series representation (2.17) of the sine function is valid for all $x \in \mathbb{R}$. None of the functions that appear in the series representation are periodic, but nevertheless the infinite sum, that is,

the sine function, is periodic with period 2π ! See Figure 2.4.7, which shows the partial sum

$$S_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!};$$

this is clearly not a 2π -periodic function. This illustrates that very contra-intuitive things might happen for infinite series; we will meet several other such instances later. $\square$

The geometric series, as well as the representations (2.15) and (2.17) derived for the exponential function and the sine function, all have the form $\sum_{n=0}^{\infty} a_n x^n$ for some coefficients a_n . In general, a series of the form $\sum_{n=0}^{\infty} a_n x^n$ is called a *power series*, and a function which can be written in the form

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

for some coefficients a_n , is said to have a *power series representation*.

Power series appear in many contexts, where one needs to approximate complicated functions. Unfortunately not all functions can be represented with help of a power series; we will come back to this on page 32.

Concerning convergence of power series, a very interesting result holds: basically, it says that a power series always is convergent on a certain symmetric interval around $x = 0$ and divergent outside the interval.

Theorem 2.4.5 *For every power series $\sum_{n=0}^{\infty} a_n x^n$ one of the following options holds:*

- (i) *The series only converges for $x = 0$.*
- (ii) *The series is absolutely convergent for all $x \in \mathbb{R}$.*
- (iii) *There exists a number $\rho > 0$ such that the series is absolutely convergent for $|x| < \rho$, and divergent for $|x| > \rho$.*

We give a proof of this result (under an extra assumption) in Section A.4. The number ρ in (iii) is called *radius of convergence* of the power series. In the case (i) we put $\rho = 0$, and in (ii), $\rho = \infty$. Theorem 2.4.5 also holds for complex values of x , which explains the word “radius of convergence”; see Figure 2.4.8, which corresponds to the case $\rho = 1$.

Example 2.4.6 In order to find the radius of convergence for the power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$, we consider $x \neq 0$ and put $a_n = \frac{x^n}{n!}$. Then

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+1}}{(n+1)!} \frac{n!}{x^n} \right| = \frac{|x|}{n+1} \rightarrow 0 \quad \text{for } n \rightarrow \infty.$$

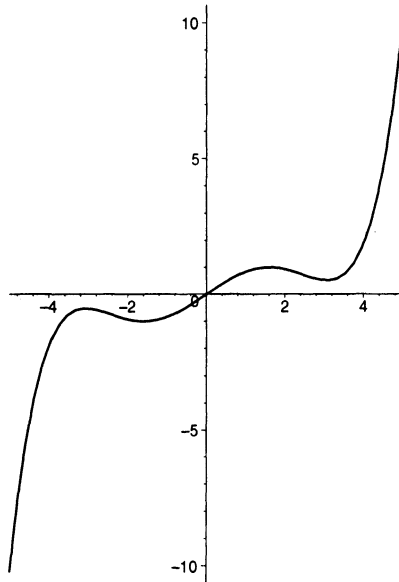


Figure 2.4.7 The partial sum $S_5(x)$ for $f(x) = \sin x$.

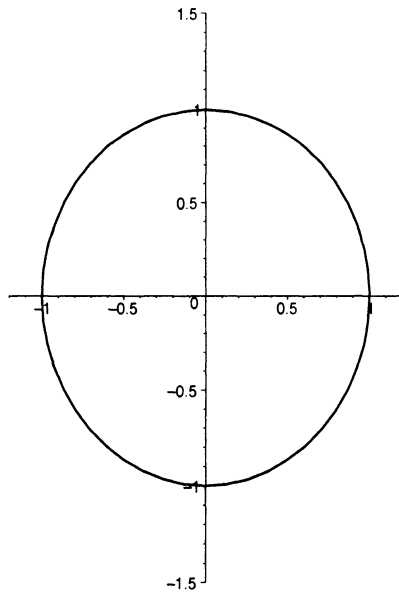


Figure 2.4.8 Circle of convergence for the series $\sum_{n=0}^{\infty} (-1)^n x^{2n}$ in Example 2.4.9, with radius of convergence $\rho = 1$. The series is convergent for x -values on the part of the real axis which is inside the circle, and divergent outside the circle. Thinking about x as a complex variable and the plane $\mathbb{R}^2$ as $\mathbb{C}$, the series is convergent for all x inside the circle.

Now the quotient test shows that the series is convergent for any value of x , so $\rho = \infty$. This is in accordance with our previous observation that (2.15) holds for all x . $\square$

Example 2.4.9 For the power series $\sum_{n=0}^{\infty} (-1)^n x^{2n}$ an argument such as in Example 2.4.6 shows that the series is convergent for $|x| < 1$ and divergent for $|x| > 1$; thus $\rho = 1$. $\square$

In the interval given by $|x| < \rho$, it turns out that a power series defines an infinitely often differentiable function:

Theorem 2.4.10 *Assume that the power series $\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence $\rho > 0$, and define the function f by*

$$f :] - \rho, \rho[\rightarrow \mathbb{C}, \quad f(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Then f is infinitely often differentiable. Moreover

$$f'(x) = \sum_{n=1}^{\infty} a_n n x^{n-1}, \quad |x| < \rho,$$

and more generally, for every $k \in \mathbb{N}$,

$$f^{(k)}(x) = \sum_{n=k}^{\infty} a_n n(n-1) \cdots (n-k+1) x^{n-k}, \quad |x| < \rho. \quad (2.19)$$

The theorem gives a part of the answer to our question about which functions have a power series expansion: in order for a function f to have such a representation, the function f must be arbitrarily often differentiable: this is a *necessary* condition. Example 2.4.12 will show that the condition is not sufficient to guarantee the desired representation. Before we present that example, we connect Theorem 2.4.10 with Taylor's theorem.

Proposition 2.4.11 *Consider a power series $\sum_{n=0}^{\infty} a_n x^n$ with radius of convergence $\rho > 0$, and let*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad x \in] - \rho, \rho[.$$

Then

$$a_n = \frac{f^{(n)}(0)}{n!}, \quad n = 0, 1, \dots, \quad (2.20)$$

i.e.,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n, \quad x \in] - \rho, \rho[.$$

Proof: First, $f(0) = a_0$. Letting $x = 0$ in Theorem 2.4.10 we get that

$$f'(0) = a_1, \quad f''(0) = 2 \cdot 1 \cdot a_2,$$

and more generally, via (2.19),

$$f^{(k)}(0) = a_k \cdot k!, \quad k \in \mathbb{N}. \quad \square$$

Proposition 2.4.11 shows that if a function f has a power series representation with some arbitrary coefficients a_n , then it is *necessarily* the representation we saw in Theorem 2.4.1. In particular, the power series representation is *unique*, i.e., only one choice of the coefficients a_n is possible. In Chapter 5 we will see examples of other types of series representations, where there are many possible ways to choose the coefficients.

As promised, we now show that not every arbitrarily often differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ has a power series representation. We do so by example, i.e., we exhibit a concrete function without such a representation:

Example 2.4.12 Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$ with the following properties:

- (i) $f(x) = 1$ for all $x \in [-1, 1]$.
- (ii) f is arbitrarily often differentiable.

Assume that f has a power series representation,

$$f(x) = \sum_{n=0}^{\infty} a_n x^n \quad \text{for all } x \in \mathbb{R}. \quad (2.21)$$

Proposition 2.4.11 shows us that

$$a_n = \frac{f^{(n)}(0)}{n!} = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Inserting this in (2.21) shows that

$$f(x) = \sum_{n=0}^{\infty} a_n x^n = 1, \quad \forall x \in \mathbb{R}.$$

It follows that unless f is a constant function, f does not have a power series representation that holds for all $x \in \mathbb{R}$. So all we have to do is to exhibit a concrete function $f : \mathbb{R} \rightarrow \mathbb{R}$, which is not constant, but satisfies (i) and (ii). One choice is the function

$$f(x) = \begin{cases} 1 & \text{for } |x| \leq 1, \\ 1 - e^{\frac{1}{x^2-1}} & \text{for } |x| > 1. \end{cases} \quad (2.22)$$

This function is shown in Figure 2.5.1. At a first glance, the function does not appear to be differentiable at $x = 1$! However, that this is indeed the

case seems more reasonable from Figure 2.5.2, which shows the function in a small neighborhood of $x = 1$.

A mathematical proof of the fact that the function is arbitrarily often differentiable falls outside the scope of this book. $\square$

The result in Example 2.4.12 can be formulated differently. In fact the example exhibits a function f for which the Taylor series $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ converges for all $x \in \mathbb{R}$, but with a sum which does not equal $f(x)$ for all $x \in \mathbb{R}$.

We notice that it is known exactly which functions can be represented via a power series: it is the class of analytic functions (i.e., the functions that are complex differentiable). More details on this can be found in any standard textbook on complex analysis, e.g. [7].

2.5 General infinite sums of functions

As we have seen, a power series consists of infinite sums of terms $a_n x^n$, $n \in \mathbb{N}$. Since not all functions have a representation of this type, it is natural also to consider infinite sums of other types of “simple” functions. More generally, we will consider a family of functions $f_0, f_1, f_2, \dots$ with the *same* domain of definition I , and attempt to define the function

$$f(x) = f_0(x) + f_1(x) + f_2(x) + \cdots + f_n(x) + \cdots = \sum_{n=0}^{\infty} f_n(x). \quad (2.23)$$

Here it is of course important to know for which values of x the expression for $f(x)$ makes sense: this is the case exactly for the $x \in I$ for which *the series of numbers* $\sum_{n=0}^{\infty} f_n(x)$ is convergent. The function f is called *the sum function* associated to $f_0, f_1, \dots$. As before we will call the expression

$$S_N(x) = f_0(x) + f_1(x) + \cdots + f_N(x)$$

the N th partial sum of $\sum_{n=0}^{\infty} f_n(x)$; it is now a function depending on x .

In practice, one can only deal with a representation of the type (2.23) if the functions f_n have some kind of common structure. In the next chapters we will consider infinite series where the functions f_n are trigonometric functions or elements from a wavelet system; both types of systems satisfy this requirement.